

Ziad Albert Matni

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ACADEMIC APPOINTMENTS

2022 – present	Assistant Teaching Professor (LPSOE), Computer Science, University of California, Santa Barbara
2016 – 2022	Lecturer , Computer Science, University of California, Santa Barbara Lecturer , Communication, University of California, Santa Barbara
2011 – 2016	Part-Time Lecturer , Information and Computer Science, School of Communication & Information (SC&I), Rutgers University
2011 – 2014	Research Fellow , SC&I, Rutgers University
1994 – 1995	Teaching Assistant , Department of Electrical Engineering, Viterbi School of Engineering, University of Southern California

EDUCATION

Ph.D.	2018	Rutgers University, School of Communication and Information (SC&I), New Brunswick, NJ. Emphasis: Information Science. Dissertation Title: <i>The Influence of Network Structures and Information Seeking Uncertainty on Information Seeking Behavior</i> . Advisor: Dr. Chirag Shah. Online link to dissertation: https://bit.ly/3GcY6Kv
M.S.	1996	University of Southern California, Viterbi School of Engineering, Los Angeles, CA Electrical Engineering, Emphasis: Communication Systems Design
B.S.	1994	University of Southern California, Viterbi School of Engineering, Los Angeles, CA Electrical & Computer Engineering, <i>Cum Laude</i>
B.A.	1994	St. Mary's College of California, Moraga, CA Liberal Arts (Physics minor), <i>Summa Cum Laude</i>

RESEARCH INTERESTS

My two main areas of research interest are Computer Science Education and Information Seeking Behavior around communication technologies via networked communities. My dissertation examined people's behavior around information seeking in various scenarios and in the context of their social network structures. My research is computational and data-science-oriented in nature and often comprises of building electronic, programmable data collection tools and capturing subjects' behavior as they interact with these tools. My pedagogical research interests extend to understanding the most effective ways that students learn computational methods using information systems, including the use of LLMs in Computer Science education.

JOURNAL ARTICLES

Appel, L., Dadlani, P., Dwyer, M., Hampton, K. N., Kitzie, V., **Matni, Z. A.**, Moore, P., & Teodoro, R. (2014). Testing the validity of social capital measures in the study of information and communication technologies. *Information, Communication & Society*, 1–19. doi:10.1080/1369118X.2014.884612

McInerney, C.R., & **Matni, Z.** (2013). Knowledge management and community and knowledge management and systems or processes: Introduction to the “Virtual Issue on Knowledge Management and Community” in the *Journal of the American Society of Information Science & Technology (JASIST)*, C.R. McInerney and Z. Matni (Eds.). [http://onlinelibrary.wiley.com/journal/10.1002/\(ISSN\)1532-2890/homepage/knowledge_management.htm](http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1532-2890/homepage/knowledge_management.htm)

PUBLISHED PROCEEDINGS

Mirza, D., Conrad, P., Lloyd, C., **Matni, Z.**, Gatin, A. (2019). Undergraduate teaching assistants in computer science: A systematic literature review. In the *Proceedings of the Fifteenth Annual ACM International Computing Education Research (ICER) conference*, Toronto, ON, Canada.

Choi, D., **Matni, Z.**, & Shah, C. (2016). What social media data should I use in my research?: A comparative analysis of Twitter, YouTube, Reddit, and the New York Times comments. In the *Proceedings of the Association for Information Science & Technology (ASIST) Annual Meeting*, November 2016, Copenhagen, Denmark.

Matni, Z., Sonne, J., & Choi, D. (2015). Society Key: Integrating social media data with governmental open data to encourage community wellbeing. In *iConference 2015 Proceedings*. Newport Beach, CA.

Choi, D., **Matni, Z.**, & Shah, C. (2015). Switching sources: A study of people’s exploratory search. In *Proceedings of Association for Information Science & Technology (ASIST) Annual Meeting*. St. Louis, MO.

Matni, Z., Choi, D., & Shah, C. (2014). A demonstration of SOCRATES: Streamlining the collection, analysis and exploration of social media data. In *Proceedings of the Association for Information Science & Technology (ASIST) Annual Meeting*, November 2014, Seattle, WA.

Matni, Z., & Shah, C. (2014). For the love of information: Motivations and affective dynamics of surfing the Web for pleasure. In *Proceedings of Association for Information Science & Technology (ASIST) Annual Meeting*. Seattle, WA.

Schwartz, R., Naaman, M., & **Matni, Z.** (2013). Making sense of cities using social media: Requirements for hyper-local data aggregation tools. In *Proceedings, WCMC Workshop at ICWSM*, July, Boston, MA.

BOOK CHAPTERS

Matni, Z., and Stohl, C. (2017). Organizational networks and strategic communication in *The International Encyclopedia of Strategic Communication*, Wiley.

CONFERENCE PRESENTATIONS

- Choi, D., **Matni, Z.**, & Shah, C. (2015). SOCRATES 2.0: Bridging the gap between researchers and social media data through natural language interactions. Poster and demo presented at the *Association for Information Science & Technology (ASIST) Annual Meeting*, November 2015, St. Louis, MO.
- Matni, Z.** (2015). Detecting measures for community wellbeing on social media. Paper presented at the *Social Media & Society Conference*, July, 2015, Toronto, Canada.
- Matni, Z.** (2015). Using social media data to measure and influence community wellbeing. Paper presented at the doctoral consortium at the *15th ACM/IEEE-CS Joint Conference on Digital Libraries (JCDL)*, June 2015, Knoxville, TN.
- Matni, Z.**, Doerfel, M., & Weber, M. (2014). Surviving disruption through resilient networks. Paper presented at the *SunBelt XXXIV Conference (INSNA)*, February 2014, St. Pete Beach, FL.
- Appel, L., Dadlani, P., Dwyer, M., Hampton, K., Kitzie, V., **Matni, Z.**, Moore, P., & Teodoro, R. (2014). Awareness of social capital during Hurricane Sandy. Paper presented at the *SunBelt XXXIV Conference (INSNA)*, February 2014, St. Pete Beach, FL.
- Hampton, K., Appel, L., Dadlani, P., Dwyer, M., Kitzie, K., **Matni, Z.**, Moore, P., & Teodoro, R. (2013). Use and misuse of scales in the study of online social capital: Testing the validity of social capital measures. Paper presented at the *National Communication Association Conference (NCA)*, November 2013, Washington, D.C.
- Appel, L., Dadlani, P., Dwyer, M., Hampton, K., Kitzie, K., **Matni, Z.**, Moore, P., & Teodoro, R. (2013). Apples to apples? The validity of scales used in the study of online social capital. Paper presented at *The American Sociological Association (ASA)*, August 2013, New York City, NY.

INVITED WORKSHOPS AND PRESENTATIONS

- Matni, Z.** (2024). The Dual Mentoring Structure of ERSP. Presented at the ACM SIGSCE Workshop on Successfully Engaging Early Undergrads in Computing Research (virtual workshop), July 10, 2024.
- Matni, Z.** (2022). Introduction to MIPS Assembly Language Programming. Presented at the Computer Science Department at the University of California, Santa Barbara, March 31, 2022. Santa Barbara, CA.
- Matni, Z.** (2021). Teaching data science to social science students. Presented at the Data Science Summit at the University of California (Fall '21) on December 3, 2021. Santa Barbara, CA
- Matni, Z.** (2019). Big data uses in social media research. Presented at the Center for Information Technology & Society (CITS) Research Computing Skills Workshop. University of California, Santa Barbara, May 30, 2019. Santa Barbara, CA.
- Matni, Z.** (2019). Scraping and garnering data for the analysis of social media. Presented at the Center for Information Technology & Society (CITS) Research Computing Skills Workshop. University of California, Santa Barbara, April 20, 2019. Santa Barbara, CA.
- Matni, Z.** (2018). Teaching recursive (fun)ctions. Presented at the Computer Science Department at the University of California, Santa Barbara, February 9, 2018. Santa Barbara, CA.

Matni, Z. (2016). Using big data to inform communication research. Presented at the Communication Department at the University of California, Santa Barbara, March 16, 2016. Santa Barbara, CA.

Matni, Z. & Albertson, K. (2014). A demonstration of the SOcial and CRowdsourced AcTivities Extraction System (SOCRATES) for use in scholarly social media research. Presented at the DIMACS Workshop on Building Communities for Transforming Social Media Research Through New Approaches for Collecting, Analyzing, and Exploring Social Media Data", Rutgers University, April 10-11, 2014. New Brunswick, NJ.

Matni, Z. (2013). Using Big Data in social media to make sense of cities. Presented at the University of Western Australia Business School, June 7, 2013. Perth, Western Australia.

INVITED TECHNICAL PUBLICATIONS (Non-Peer Reviewed)

Matni, Z. (2021). Social Network Analysis, *Information Matters*, Vol. 1, Issue 8, August 2021, <https://r7q.22f.myftpupload.com/2021/08/social-network-analysis-part-1/> and <https://r7q.22f.myftpupload.com/2021/08/social-network-analysis-part-2/>

Matni, Z. (2004). When good eyes go bad: Perfecting measurement for high-speed integrated circuits, *Test & Measurement World*, April 2004, <http://www.tmworld.com/eyes>

Matni, Z. (2002). Demonstrating a fiber optic link for 10 Gigabit-per-second data communications. White paper for Inphi Corporation, December 2002.

ACADEMIC AWARDS AND FUNDING

- Outstanding Faculty Member in Computer Science Award, University of California, Santa Barbara, 2025.
- Dissertation Support grant, School of Communication and Information, Rutgers University, 2017 (\$1000).
- Distinguished Achievement as a Part-Time Lecturer Award, School of Communication and Information, Rutgers University, 2016.
- Travel and conference funding from Rutgers SC&I to participate in and attend the Association for Information Science & Technology (ASIS&T), St. Louis, MO, 2015 (\$900).
- Travel and conference funding from Rutgers SC&I to participate in and attend the Social Media & Society Conference, Toronto, Canada, 2015 (\$700).
- Travel and conference funding from the Association for Computing Machinery (ACM) to participate in and attend the 15th ACM/IEEE-CS Joint Conference on Digital Libraries (JCDL) at their Doctoral Consortium, Knoxville, TN, 2015 (\$750).
- SC&I Scholarship Award, School of Communication & Information, Rutgers University, Spring 2015 (\$700).
- Social Media Expo Team Finalist Award from the iConference 2015, Newport Beach, CA (\$3,000).
- SC&I Academic Fellowship Award, School of Communication & Information, Rutgers University, 2011– 2014 (\$25,000 per year for 3 years with full tuition remission). This fellowship is awarded to only one person per incoming cohort at Rutgers SC&I.
- Outstanding Research Practicum Display Award, School of Communication and Information, Rutgers University, 2014.
- Travel and conference funding from Rutgers SC&I to participate in and attend the Association for Information Science & Technology (ASIS&T), Seattle, WA, 2014 (\$1000).

- Travel and conference funding from Rutgers SC&I to participate in and attend the SunBelt XXXIV Conference (INSNA), St. Pete Beach, FL, 2014 (\$700).
- Outstanding Teaching Assistant Award, School of Electrical Engineering, USC, 1995.

TEACHING (Instructor of Record)

University of California, Santa Barbara, Department of Computer Science, 2016 – present

CMPSC 8*	Introduction to Computer Science	2017 – present
CMPSC W8*	Introduction to Computer Science, Online Course	2022 – present
CMPSC 16*	Problem Solving with Computers, I	2016 – present
CMPSC 32*	Object Oriented Design and Implementation	2022 – present
CMPSC 40	Foundations of Computer Science	2023 – present
CMPSC 64	Computer Organization and Design Logic	2017 – present
CMPSC 110	Research Methods in Computer Science	2023 – present
CMPSC 111	Introduction to Computational Science	2019 – present
CMPSC 154	Computer Architecture	2020 – present
CMPSC 176A	Introduction to Computer Communication Networks	2021 – present
CMPSC 193	Internship in Industry	2018 – present

University of California, Santa Barbara, College of Creative Studies – Computing, 2025

CMPTGCS 2	Foundations of Computer Science (CCS-Computing)	2025
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University of California, Santa Barbara, Department of Communication, 2016 – 2022

COMM 88*	Communication Research Methods	2017 – 2022
COMM 169	Social Networks	2016 – 2022
COMM 187^	Data Science in Communication	2020 – 2022
COMM 199RA	Independent Study (Directed Research)	2017 – 2022

Rutgers University, 2010 – 2016

ITI 201	Intro. to Computer Programming Concepts using Java	2015
ITI 331	Networking & Internet Technologies	2010 - 2016
MLIS 550	Info. Technologies for Information Agencies (masters-level)	2010 - 2016
ITI 210	Management of Technological Organizations	2012
ITI 220	Retrieving & Evaluating Electronic Information	2011
ECE 411	Electrical Energy Conversion	2010

The College of New Jersey, 2011

ENG 212	Circuits Analysis	2011
ENG 214	Circuits Analysis Laboratory	2011
ELE 333	Electrical Engineering Laboratory I	2011

University of Southern California, School of Engineering (Teaching Assistant)

EE 220	Electronic Circuits	1994 – 96
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* indicates classes that are routinely very large size (150-300 students)

^ indicates courses that I designed and initiated

Teaching Accomplishments

- Have taught over **7,180 students at UCSB** since Fall 2016.
- Have taught over **8,400** students in my teaching career since 2011.

SERVICE

To the Department

Teaching and Mentoring Students

- Director, Early Research Scholars Program (ERSP), CS Department, 2023 – present.
- Co-director, Undergraduate Learning Assistant (ULA) program, CS Department, 2022 – present.
- Co-chair/member for **7** graduate students at UCSB (1 in Communication, 6 in CS).
- Have written over **860** letters of recommendation to **280** UCSB students since Fall 2017.
- Volunteer mentor for the Women in Computer Science (WiCS) ‘Bigs and Littles’ initiative, 2023.
- Advise and mentored **23** students in Computer Science through their industry internships via CS 193.
- Lead bi-annual industry internship panel for the CS undergraduate students through the Undergraduate Affairs Committee, 2022 – present.
- Advisor/co-advisor, ACM Student Chapter, SB Hacks, Coders SB, Data Science Club.
- Regularly offer a “Basic Linux Use” workshop at UCSB for CS undergraduate students.
- Developed “Data Science in Communication Research” course, Communication Department (COMM 187).
- Created and presented workshops on Big Data methodologies and techniques for graduate students (targeted to students in the Social Sciences), Center for Information Technology and Society (CITS), UCSB, April/May 2019.
- Taught **17** students computational and programming methods for Comm. research (COMM 199RA).

Graduate Student Committees – UCSB

- Yoori Yang, 6/2021, Member, Communication PhD dissertation committee
- Ashley Wang, 6/2023, Co-Chair, MS Computer Science thesis committee
- Rory Zahedi, 6/2023, Co-Chair, MS Computer Science thesis committee
- Zechen Ma, 6/2024, Co-Chair, MS Computer Science thesis committee
- Ethan Meade, 6/2024, Co-Chair, MS Computer Science thesis committee
- Jiajun Wang, 3/2025, Co-Chair, MS Computer Science thesis committee
- Guangzhe Liu, 6/2025, Co-Chair, MS Computer Science thesis committee

Department Committees - UCSB

- Chair, Course Assessment Task Force (CS 16/24/32), CS Department, UCSB, 2024.
- Chair, Course Assessment Task Force (CS 176A/B/C, CS 171), CS Department, UCSB, 2023.
- Member, Undergraduate Affairs Committee, CS Department, UCSB, 2022 – present.
- Member, Course Assessment Task Force (for CS 8/9/16), CS Department, UCSB, 2021.
- Member, Course Assessment Task Force (for CS 111/165A/165B), CS Department, UCSB, 2020.
- Member, Undergraduate Committee, Communication Department, UCSB, 2017 – 22.
- Managing Editor, Communication Department Newsletter, UCSB 2019 – 22.

Department Committees – Rutgers University

- Panelist, School of Communication and Information doctoral program invitational, 2015 – 16.
- Panelist, SC&I, Advisory Council Meeting, 2015.
- Webmaster, Doctoral Student Association (DSA), SC&I, Rutgers University, 2014 – 15.
- Panelist, SC&I doctoral program invitational, 2014.
- Panelist, DSA new student orientation, 2012 – 2014.
- Treasurer, Doctoral Student Association (DSA), SC&I, Rutgers University, 2012 – 13.

To the University

- Invited faculty member for the UCSB chapter of WiCS receptions, 2021 – 25.
- Invited faculty judge for the UCSB SB Hacks for 7 consecutive years, 2019 – 25.
- Invited faculty judge for the UCSB Coders SB, 2023.
- Invited faculty judge for Data Science UCSB, 2025.
- Invited faculty mentor for the UCSB Regents Scholars reception, 2019 – 22.
- Member, University Career Development Awards Committee, 2025.
- Member, University Committee for Faculty Welfare (CFW), UCSB 2020.

To the Discipline

Editorship

- Co-editor, “Virtual Issue on Knowledge Management and Community” for the Journal of the American Society of Information Science & Technology (JASIST), 2013.
[http://onlinelibrary.wiley.com/journal/10.1002/\(ISSN\)1532-2890/homepage/knowledge_management.htm](http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1532-2890/homepage/knowledge_management.htm)

Professional Affiliations and Conferences

- Treasurer, New Jersey Chapter of the Association for Information Science and Technology (NJASIST), 2014 – 16. Administer and manage all financial needs (approx. \$20,000 in assets) for the state-wide chapter of an international organization.
- Treasurer, Rutgers University Chapter of the Association for Information Science and Technology (RUASIST), 2014 – 16.
- Co-organizer, "DIMACS Workshop on Building Communities for Transforming Social Media Research Through New Approaches for Collecting, Analyzing, and Exploring Social Media Data", Rutgers University, April 10-11, 2014.

PROFESSIONAL AFFILIATIONS

- Member, ACM, 2013 – present.
- Member, IEEE, 1996 – present.
- Member, ASIS&T, 2013 – present.
- Member, Rutgers Information Seeking Lab (Dr. Chirag Shah, director), 2013 – 16.
- Member, Network SCI Lab (Dr. Marya Doerfel and Dr. Matthew Weber, directors), 2013 – 16.
- Member, Rutgers Social Media Information Lab (Dr. Mor Naaman, director), 2011 – 13.

PROFESSIONAL AND TECHNICAL EXPERTISE

RESEARCH METHODOLOGIES USED

- Computational modeling
- Quantitative analysis, multivariate statistical analysis
- Big Data and Data Science methods, especially for use in the Computational Social Sciences
- Social Network Analysis (SNA)
- Qualitative analysis, including interviews, content analysis, focus groups, surveys, etc...

TECHNICAL EXPERTISE

- Advanced skills in programming/scripting languages: Python, C/C++, Java, MIPS assembly, UNIX csh/bash, MATLAB, Perl, PHP, HTML, CSS, and JavaScript.
- Hardware simulation languages: HSPICE (analog), Verilog, VHDL, pyRTL (digital).
- Social network analysis platforms: R Statnet, UCInet.
- Statistical analysis platforms: R, SPSS.
- Large data set mining techniques for Big Data: web-scraping (BeautifulSoup), API programming.
- Crowdsourcing tools: Amazon Mechanical Turk.
- Relational database management systems: MySQL, Mongo DB.
- Amazon Web Services: Elastic Compute Cloud (EC2), Simple Storage Service (S3), Relational Database Service (RDS).
- Network server design and system administration: RAID servers, UNIX system administration.
- Computer network system design and management/administration.
- Human-Computer Interface (HCI) user-experience (UX) design.

OTHER PROFESSIONAL SKILLS / RELEVANT INFORMATION

- Design engineering of FPGA, System-on-Chip, and custom semiconductor integrated circuits.
- Design engineering of electronic and optoelectronic network communication systems.
- Marketing and management: Product and project management for computer communication systems.
- Fully fluent in English, French, and Arabic.
- U.S. citizen.

INDUSTRY EMPLOYMENT HISTORY

Matni Consulting

2009 – 2010

- Self-employed: Consulted with various investment banks on the state of the Semiconductor Industry.

LSI Corporation (previously Agere), Allentown, PA

2005 – 2009

Product Marketing Manager

- Drove new product definition, product prioritization, competitive analysis, and communication across business unit for state-of-the-art digital data storage devices.

Red Sky Systems, South Plainfield, NJ

2005

Senior Hardware Engineer

- Engineering lead on designs for an undersea optical fiber data communication system. Ensured reliability and manufacturability based on stringent commercial, industrial and military specifications.

Inphi Corporation, Westlake Village, CA

2002 – 2005

Applications Engineering Manager

- Managed over 100 products for customer engineering and marketing support for state-of-the-art ultra-high-speed electronic communication devices.
- Drove client support via product presentations, white papers, design guides, trade journal articles, and ensured compliance with multiple industry standards.

Vitesse Semiconductor Corporation, Camarillo, CA

1996 – 2002

Design Engineer, Senior Applications Engineer

- Designed multiple complete integrated circuits (IC) for high-speed communication systems in a Gallium Arsenide (HGAs-4) semiconductor fabrication process.